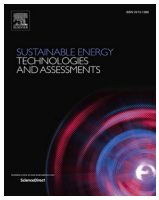
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Review article



Model predictive control in real-world buildings – a systematic review of implementation and performance

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ABSTRACT

This paper investigates the application of Model Predictive Control (MPC) for optimisation of building operation, with a focus on implementation and scalability. A systematic review was conducted to categorise MPC application areas and prediction models, identify key challenges and limitations, and evaluate the overall performance reported for MPC. The results show that MPC implementations vary considerably with respect to adopted methodology, climatic conditions and control objectives. Reported energy savings range from a few per cent up to 56.3%, while cost savings reach up to 55%. MPC applications are most prevalent in hydronic heating systems due to their flexibility in integrating multiple energy sources. In smart-city contexts, MPC supports energy efficient design and on-site renewable energy generation by improving self-consumption and enabling load shifting. Grey-box models are the most widely used modelling approach, accounting for 62% of implementations overall and 86% in residential single-family buildings, where IoT devices and smart sensors are frequently utilised. Despite these advances, limited commercialisation remains a key barrier, highlighting the need for further research on scalable deployment and market adoption.

Introduction

The vast majority of the scientific community has reached a broad consensus that human induced global warming presents a critical threat requiring urgent action. Several reports from the United Nations Environment Programme (UNEP) and the Intergovernmental Panel on Climate Change (IPCC) emphasise the need for rapid reductions in greenhouse gas (GHG) emissions [1–3]. Achieving these reductions necessitates coordinated action across multiple sectors. This has increased the focus on digitalisation, predictive approach, and sustainability oriented operational strategies [4–6].

The buildings sector is of particular importance in this context. According to the International Energy Agency (IEA), buildings account for approximately 30% of global final energy consumption and 26% of energy related CO_2 emissions [7]. An effective strategy for reducing GHG emissions in buildings involves optimisation of building operational performance in conjunction with locally generated and grid connected renewable energy systems. Within this framework, MPC has emerged as a particularly promising control methodology. This is due to its ability to systematically incorporate system dynamics, operational constraints, and forecasts of energy demand and local renewable energy generation into the decision-making process. It is also worth mentioning that beyond buildings, MPC is long standing and widely deployed in process industries, power electronics, transport, and robotics [8–11].

MPC relies on dynamic system models and forecasts to determine control actions. In building operation this capability is used to reduce energy consumption and operational costs while at the same time preserving occupant comfort. Numerous simulation studies, along with several pilot implementations, have reported energy savings in the range of approximately 10–30% when MPC is utilised in buildings [12–16]. Recent advances in MPC include robust and stochastic formulations, economic MPC, and distributed or hierarchical architectures. These are complemented by learning-based methods and faster embedded solvers. Together, these developments have broadened the practical capabilities of MPC [11,17–20]. These developments are directly relevant to buildings, where forecast uncertainty, multi-zone coupling, and explicit comfort and actuator constraints are central. Nevertheless, a gap persists between promising results and routine deployment in buildings, highlighting the need for systematic assessments of implementation and scalability.

Generally, MPC setup includes pre-assessment, a control-oriented model, an MPC controller and the controlled system. The control-

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oriented model predicts future building states, and is typically categorised as *white-box*, *black-box* and *grey-box*. *White-box* models describe system dynamics from physical laws and known relationships, such as the laws of thermodynamics. *Black-box* models learn the dynamics from historical and measured data without explicit physical relationships. *Grey-box* models combine both, utilising simplified physics with parameters estimated from data. A common grey-box approach is the resistance–capacitance (RC) model, which represents a building's thermal behaviour using an electrical circuit analogy where resistors and capacitors correspond to thermal resistances and capacitances. While the model predicts how the building responds to inputs and disturbances, the controller adjusts the system to optimise one or more objectives. In buildings, this usually involves minimising energy use while ensuring thermal comfort. Objectives are encoded in an objective function, and an optimiser computes control actions that yield the best operation given the model's predictions and constraints. A typical MPC setup for building applications is shown in Fig. 1. The aim of this review is not to provide an in depth account of MPC, but to address methods proposed for implementing predictive control techniques. Accordingly, only the fundamentals necessary for the review are presented, and detailed explanations are omitted. Comprehensive reviews of MPC applications in buildings are provided in [12,14].

Several review articles have surveyed the broader field of MPC in buildings, providing insight into both challenges and opportunities. For example, one review investigated optimised control systems with a particular emphasis on simulation tools [21]. It highlighted the increasing adoption of automated control and optimisation strategies, their potential to reduce GHG emissions, and the critical role of occupant behaviour and preferences in achieving effective and reliable automation. Another review proposed a unified framework for MPC with an emphasis on real-world applications [12]. It covered six themes, namely MPC theory and problem classes, algorithmic solutions, software tools for building modelling and control, practical deployment in buildings, performance assessment, and market prospects. The review sought to build a common understanding between researchers and practitioners. Despite uncertainties, the authors argued that MPC could achieve large-scale market penetration within the next decade through customised algorithms delivered via contemporary building automation systems (BAS) or through integration into software platforms.

A review of grey-box modelling for building energy simulations identified five main challenges, namely theoretical limitations, ambiguous structure and terminology, vague creation processes, unclear application domains, and the absence of unified software solutions for wider adoption [22]. A recent science mapping study adopted a quantitative approach and showed that most publications on MPC in buildings still focus on simulation and optimisation rather than field deployment [23]. It also set out research priorities, including validating model complexity against accuracy, and improving sensing data analysis and labelling to enable broader operational adoption.

Only a few studies have summarised MPC field implementations in buildings, cataloguing 12, 14 and 24 cases, respectively [15,24,25].

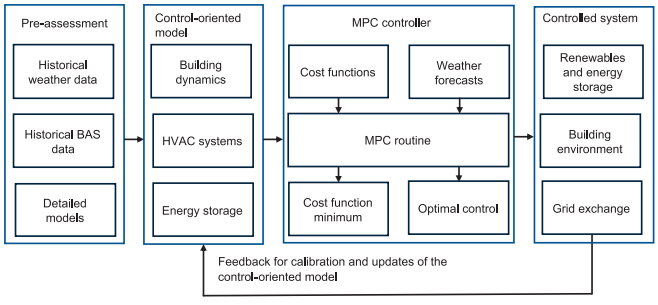


Fig. 1. Schematic representation of a typical MPC setup for building applications.

These summaries focus on specific cases without systematic cross comparison of how MPC was implemented, what operational performance was achieved in operation, and which factors support scalability across building types and energy systems. Moreover, studies involving buildings with MPC operation remain fragmented, and it remains unclear which design choices and data requirements drive operational performance. Accordingly, a comprehensive and systematic synthesis of operational MPC implementations is still lacking. Such a synthesis should compare modelling choices, sensing and forecasting requirements, control formulations, and transferable practices. To address these gaps, a systematic literature review of real-world MPC implementations in buildings is presented, focusing on studies published between 2020 and 2024. The main contributions are to:

Map recent MPC studies and identify those with operational implementations, classifying studies by modelling choice, building type, geography, control objectives, energy carriers and HVAC scope.

Characterise common factors in the reviewed implementations, and identify how context, control design and configuration relate to reported outcomes including energy and cost savings, comfort and flexibility.

Provide practical suggestions for scalable deployment and transferable practices across diverse studies and contexts.

The results are organised by implementation features, contexts, models, sensing and data, scalability and performance. Focusing on operational buildings, the review complements theory led surveys and offers practical pointers on incentives, interoperability and straightforward measurement and verification. The remainder is structured as follows. Section 2 describes the methods, including data collection and evaluation procedures. Section 3 presents the outcomes of the bibliographic and content analysis, while Sections 4 and 5 discuss the main findings and the outlook.

Method

Research on MPC for building energy management has grown substantially in recent years. For example, a Scopus search for "MPC OR Model predictive control AND building OR buildings" returned 5,490 records on 24 September 2024. Annual outputs have grown from only a handful in the early 2000 s to several hundred in recent years. To examine the breadth of this research, data from the Scopus search was exported and analysed using VOSviewer v.1.6.19 [26], focusing on the co-occurrence of keywords. Selecting keywords that occurred more than 50 times yielded 232 results, which are presented in Fig. 2 as a network visualisation. Upon reviewing these results, it was found that there was a notable absence of terms related to practical implementations, such as "implementation" "field studies" and "scalability", indicating that this has not been in focus of most of the research.

A systematic approach to keyword synonym searching was used, and a final search string was defined to establish the boundaries and objectives of the review. The aim was to capture all articles relevant to the review scope, focusing on (1) implementation and scalability, particularly the real-world application of (2) MPC in (3) building automation systems. These three aspects were defined as search blocks, each combined using an AND operator in the search string. Fig. 3 presents a flow chart illustrating the selection process for the string search, where the search objective was predefined.

For each block, synonyms and related words were used to capture all pertinent articles, where the initial search string included several synonyms for different terms. After refining the string according to Fig. 3, it became apparent that reducing the number of synonyms did not decrease the number of relevant articles. This suggests that terminology in this domain is relatively standardised.

Search terms related to thermal comfort and indoor air quality (IAQ) did not yield significantly more hits, likely because heating or cooling is usually investigated when considering these aspects. Consequently, the final search string is relatively straightforward, as shown in, top-right corner in Fig. 3. In addition, two exclusion terms, "phosphate" and

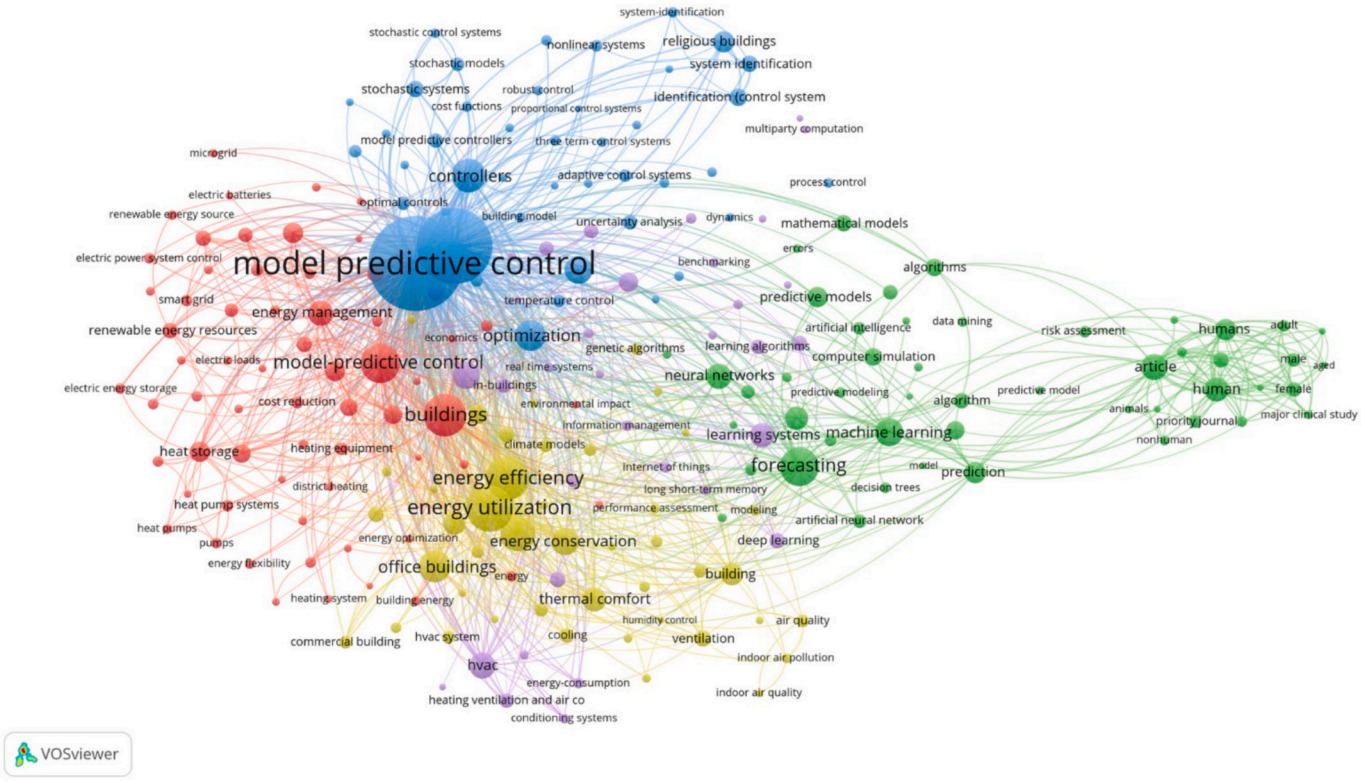


Fig. 2. Network visualisation of keywords occurring more than 50 times out of the 5,490 hits.

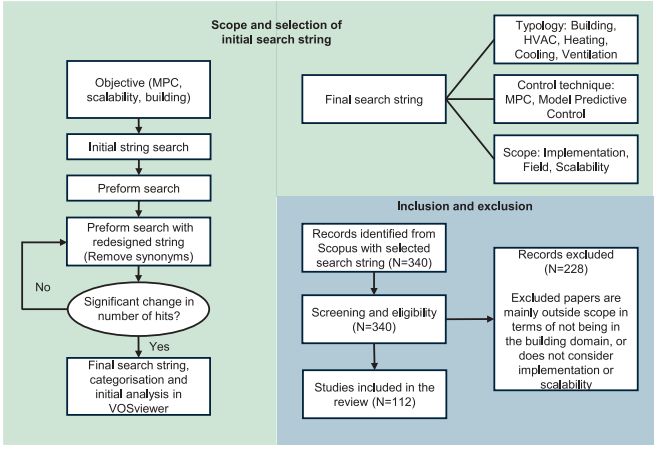


Fig. 3. Flowchart of the search strategy, including selection of the initial search string and study inclusion and exclusion criteria.

"magnesium," were included to avoid irrelevant results related to Magnesium Phosphate Cement, abbreviated as MPC.

Scopus was chosen as the sole database for the literature search due to its comprehensive coverage and inclusion of high quality, peer-reviewed sources. Moreover, Scopus provides robust citation tracking, which allows for detailed analysis in tools like VOSviewer. Utilising a single database enhances the reproducibility of the results and given that the majority of relevant literature is indexed in Scopus, it was deemed the most suitable choice for this review.

Given that most articles on MPC in buildings have been published in recent years, the review focuses on research conducted between 2020 and 2024 to highlight the latest advancements and trends in the field. Since implementation typically represents a later stage in the research lifecycle, following extensive theoretical, experimental, and simulation work, restricting the scope to the past four years ensures a focused and relevant dataset. This approach results in data that are both timely and comprehensive.

Using the selected search string and constraints, 340 articles were found. Each of these articles was screened manually, excluding articles that are not relevant to the objective. Some articles could easily be excluded by clearly not fitting the scope (e.g., heating of the cockpit of a car), while others were more difficult to assess. For example, simulation studies that include considerations about scalability and real-world implementation posed a challenge. In these cases, inclusion decisions were made based on the review objectives. After manually scanning all hits, a total of 112 articles was included in this review paper. It is worth noting that the discrepancy between the total number of references and the included articles is due to references in the Introduction that fall outside the scope and are therefore not analysed in detail.

Together, the three block search strategy with explicit exclusions and the year window define a transparent, reproducible screening and coding protocol. Additionally, with the operational definition of full-scale implementation presented in Section 3.1, this protocol is the basis for the results and statistics reported in the following section. The following section reports the screened evidence, organised from implementation scope and context to modelling choices, data requirements and outcomes.

Results and discussion

The diversity within the building stock presents varying objectives, challenges, and opportunities for MPC. This section examines factors that influence the implementation and operation of MPC in the included studies. By recognising the interplay of these factors, this study aims to identify key success factors. First, the implementation scope of the included studies is presented, providing important context for the results of the reviewed work. Following that, main factors that affect each study is presented, including geographical location, building type, energy

systems, models being used and data management and sensors. Each section aims to provide insights into how these variables impact the studies and feasibility. To support comparison across diverse cases, the synthesis proceeds from implementation features to context factors, then to models, data and reported performance.

Implementation features

As previously stated, this review includes articles that address the aspects of implementation and scalability, regardless of whether the study itself involved the direct implementation of MPC. Some case studies feature full-scale implementation in operational buildings. Others investigate MPC in parts of a building, in laboratory environments, or via simulations that address implementation aspects. Fig. 4 shows the distribution of studies according to their level of implementation.

Prior studies typically distinguish a validation ladder from simulation to testbeds or emulation to field trials or real-building operation [12,14,27]. The terminology of the different stages is typically given by intuitive wording, such as "simulation", "implementation", "operating building" and related terms. However, these labels are not applied consistently across studies and rarely make explicit whether control is closed-loop, what spatial extent is covered, or how much of the building system is under MPC. For instance, "operating building" does not indicate anything about the extent of the building space covered, since a study may target only a single room.

To reflect current practice and enable consistent comparison across studies, we adopt an operational definition of "full-scale implementation" aligned with the real building operation tier and inclusive of subsystem scope and occupancy status. In this review, "full-scale implementation" refers to studies in which MPC is applied in closed-loop to at least one subsystem, such as domestic hot water, heating, or cooling, within a real, operational building. The definition does not require control of all building energy systems and includes partial scope cases when the controller acts at building scale, meaning beyond a single room or zone. Unoccupied buildings are included if these criteria are met, which applies to two studies [28,29]. Using this definition, we identify 75 implementation studies, of which 34 meet the full-scale criterion. The remaining included articles comprise simulation studies and two review articles [23,30], all of which focus on implementation or scalability.

Full-scale implementation

To gain a deeper understanding about practical challenges, it is valuable to review studies on operational MPC in real-world environments. This helps identify issues that may occur throughout the entire process of implementing MPC. To give some context of the included studies, building type of the studies that have conducted full-scale implementation is shown in Fig. 5.

MPC in single family homes commonly integrates IoT, notably smart thermostats, to optimise energy use. Jiang [28] tested MPC with smart thermostats, node sensors and smart meters in five occupied homes and one laboratory house in the United States. Focusing on optimal pre-cooling, the study reported energy cost savings of 28.72%–51.31% on hot summer days, and −14.56% to 60.32% on mild days. The single negative value occurred in one home and was attributed to changes in the homeowner's preferred thermal comfort range. Savings depended strongly on floor area, with larger homes offering more flexibility, and on comfort preferences, highlighting the complexity of evaluating real-world performance.

Smart thermostats were also used in [15], where MPC regulated heating in a multi source system in an occupied dwelling, cutting energy use by 18.7%. A subsequent study introduced humidity aware MPC for residential cooling, improving comfort and efficiency [31]. These two studies utilise a similar supervisory control scheme. A local MPC service running on a PC with MATLAB retrieves weather forecasts via an API and recent building measurements from a cloud database. The MPC optimiser uses a control-oriented building model that includes the HVAC system, the building envelope, and internal gains to calculate temperature setpoints. These setpoints are then sent to the smart thermostat, which preforms low level control within the dwelling. This architecture exemplifies the typical integration of smart thermostats in real-world MPC implementations.

In Denmark, economic MPC enabled load shifting between hydronic radiators and domestic hot water in a single family house [32], using room sensors and customised actuators with district heating.

IoT was applied in Mauritius to control portable air conditioners and fans to cut cooling demand [33]. Although savings were not reported, the system successfully controlled a two zone unoccupied experimental building. In Canada, MPC in an unoccupied multi-zone house with smart thermostats reduced electricity costs by 55% [29]. Another study increased PV self-consumption while optimising heat pump operation in an unoccupied dwelling [34].

Despite promising results, moving from conventional control to MPC in homes remains challenging. Interoperability across smart devices is a major hurdle, for example thermostats paired with heat pumps constrained by closed manufacturer systems [15] and radiator thermostats that do not align with room sensors [32]. Additional issues include battery life, internet interruptions and low resolution energy data. Model inaccuracies and uncertain user behaviour impede field performance. Evidence from a 12 home DHW study [35] and a Canadian deployment in a highly glazed house [36] suggests that modelling errors in key disturbances (tank thermal dynamics and passive solar gains, respectively) can materially affect field performance and lead to periods of overheating or overcooling. Scalability concerns are detailed in [37], which lists 24 challenges, with protocol limitations and limited access to room setpoints from installed regulators noted as key barriers. Some

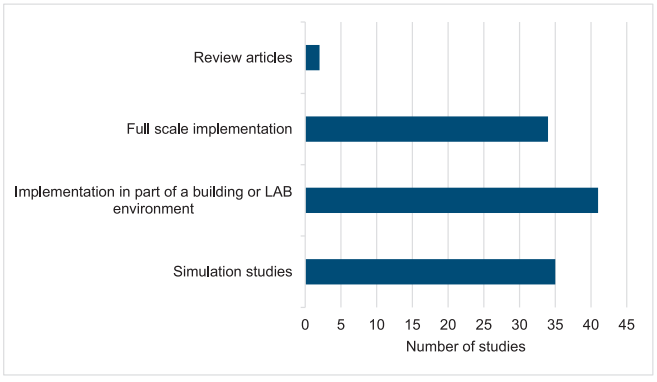


Fig. 4. Distribution of included studies according to their level of implementation.

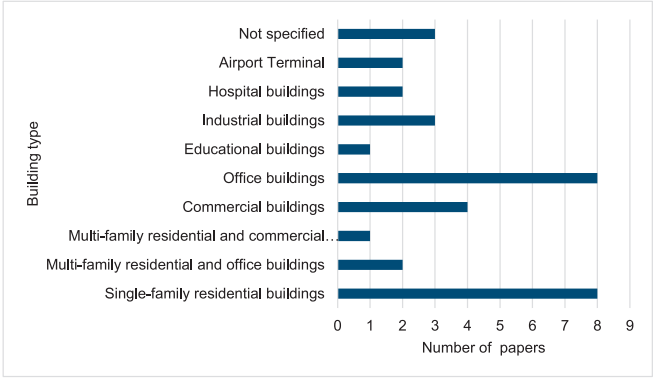


Fig. 5. Distribution of building types in full-scale implementation cases.

problems are tractable, such as correcting thermostat air temperature bias through recalibration or advanced sensing [38]. In practice, these issues often lead to modest or hard to quantify outcomes, which are sometimes reported only qualitatively, even when the controller is successfully deployed.

MPC also applies to larger residential and commercial buildings. In Poland, forecast based heating control was deployed in existing and new hydronic buildings [39]. using a simplified model driven by weather forecasts [40]. It regulated heating across multiple public and residential blocks, with average savings of 13.4% in residential and 10.7% in public buildings [41]. The method was limited to hydronic space heating with an outdoor weather regulator, so potential savings are likely lower than with advanced MPC. A comparable Polish study is reported in [42].

MPC was also deployed cost effectively in a Norwegian commercial building [43]. The study used an RC model and treated the ventilation extract temperature as a proxy for indoor temperature. Over one month in early spring, the controller reduced energy costs by 33%. Compared with [39], the RC model included more building specific parameters.

Some studies emphasise cost-effective solutions with minimal hardware, while others demonstrate large-scale MPC implementations. A notable example is a hospital in Italy, where MPC was integrated for combined heating and cooling control within a multi energy infrastructure comprising CHP, natural gas boilers, steam generators, absorption chillers and electric chillers [44]. The deployment produced a modest 0.15% reduction in total energy use but a 21.5% reduction in electricity purchases from the grid. The controller used both long term and short term predictions. The long term module captured operational goals and expected conditions and set feasible boundaries for the short term module, which computed the control actions. To realise the benefits of the long term layer, the energy system must be sufficiently flexible while maintaining comfort requirements. These results add to evidence of full-scale MPC in operating buildings. Table 1 summarises the reviewed studies, although each case is unique in scope, equipment, energy carriers and model development.

Implementation in parts of a building, single room and lab environment

Many studies on MPC have focused on implementations within parts of buildings, targeting specific rooms or controlled laboratory environments for testing and evaluation. This approach minimises risks to occupant comfort and indoor air quality while allowing researchers to assess MPC performance under controlled conditions. However, such setups often overlook actual occupancy patterns and interactions between different building zones.

As an example of such controlled evaluations, federated cooling control was implemented in a test chamber with a heat pump and photovoltaic system in South Korea, demonstrating a 40% reduction in energy consumption [63]. Similarly, an IoT equipped MPC was tested in a controlled environment in the UK, yielding a 35% reduction in electricity costs [24]. In Italy, a weather forecast based control system for phase change material (PCM) enhanced radiant floors demonstrated improved heating and cooling efficiency in a small experimental building [64]. A prototype HVAC control system in a student laboratory utilised scenario based MPC to adjust the temperature and airflow for AC and ventilation, thereby demonstrating the practical application of advanced MPC strategies [65]. Additionally, a low cost grey-box MPC implementation in South Korea used Arduino based control in a single room, achieving a 5.8% cost saving compared to traditional rule-based control [66].

Simulation studies

Simulation studies could provide valuable insights into the effectiveness of MPC strategies before field deployment, highlighting their potential for energy savings and improved performance. In our work, a particular attention has been undertaken to focus on simulation studies that explicitly address scalability or implementation. Research in this domain encompasses a wide range of control applications, including multi source hydronic heating systems and the integration of MPC with local district heating networks [67–74].

For example, a simulation study in Germany employed a white-box modelling approach of the thermal electrical renewable energy system in four different single family house standards. This system consists of several energy sources and carriers, such as solar thermal, gas boilers, DHW with thermal storage, CHP, PV and batteries. The study demonstrated up to 30% space heating demand reduction [72]. The authors have concluded that universally implementing the MPC scheme to fit more buildings requires changes in the objective function, penalties, weighting factors, and the model. They proposed an adaptive approach integrating AI driven forecasting, user customization, and initial questionnaires to tailor MPC configurations for diverse building typologies. A different study introduced a hybrid Cyber Physical System simulation and implementation to assess distributed MPC strategies for heating and cooling within a six story building comprising 40 rooms [75]. The simulation demonstrated promising results in terms of temperature control and distribution across the different rooms of the building, although energy consumption metrics were not explicitly presented. A Canadian simulation and analytical study proposed models for MPC based heating control for residential building stock [76]. The studies underscore the feasibility of extending MPC applications to broader large scale building stocks.

Other simulation studies address hydronic HVAC, radiant and convective systems, Thermal Energy Storage (TES) and active envelope measures, and district-heating operation across multiple building types. Reported outcomes include 10–30% energy or cost reductions in most cases, up to ~ 65% energy savings in specific commercial scenarios, approximately 15% peak shaving, and substantial load shifting where TES is used [77–94].

Geographical location and building type

The implementation features discussed in Section 3.1 influence how the studies are conducted. In this section, the study highlights additional factors that may affect MPC operation in the targeted case buildings, namely geographical location and building type.

For geographical location, several aspects can influence both the execution and outcomes of studies, including climate, building regulations, standards, and regional practices. Among the 112 reviewed articles, studies span 31 countries, primarily from North America, Europe, China, and South Korea, reflecting the diverse geographic contexts of MPC research. A total of 69 articles originate from Europe, seven of which are from Scandinavia, making Europe the most represented continent. A complete overview of the articles and their countries of origin is provided in Appendix I.

Building type can also influence load profiles, occupancy patterns, and system complexity, affecting the size and operational dynamics of buildings. The included studies cover a range of building types, including residential buildings (single family homes and larger apartment complexes), public buildings (educational and healthcare facilities), commercial buildings (convenience stores and shopping malls), and industrial buildings (factories and industrial greenhouses). Table 2 provides a detailed breakdown of the building types and the number of studies associated with each. Following this, examples of the most represented building types are presented.

Offices and multipurpose buildings

Several studies have focused on implementing MPC in office and multipurpose buildings. For example, a field experiment in Switzerland demonstrated a grey-box MPC for a mixed purpose building using GSHP and district heating. This approach allowed for 13.4% flexibility in electricity consumption due to the buffer provided by water storage tanks [95]. In Germany, a section of a large, low energy office building was controlled with MPC over three months. The building's energy system includes district heating, GSHP, and Thermally activated

Table 1
Summary of full-scale implementation studies.

Article	Building Type	Country	Area of Application	Energy system	Energy distribution	Building Model Method	Savings
[34]	Residential (single family house)	Germany	Heating, load shifting	Heat pump, PV, battery storage	Hydronic	Black-box	36% decrease in grid energy (Higher self-sustainability)
[33]	Residential (single family house)	Mauritius	Cooling, heating	Portable AC, extractors	Forced air	Black-box	Not specified
[31]	Residential (single family house)	USA	Cooling, Load shifting	Heat pump	Forced air	Grey-box	27% annual cooling and heating energy cost reduction
[15]	Residential (single family house)	USA	Heating	Air-to-air heat pump with electric resistance backup	Forced air	Grey-box	Average reduction of 19% heating energy
[28]	Residential (single family house)	USA	Cooling	AC	Forced air	Grey-box	28–51% energy cost savings
[29]	Residential (single family house)	Canada	Heating and load shifting	Electric panels	Electric heating system	Grey-box	Average cost of electricity reduced by 55%
[36]	Residential (single family house)	Canada	Heating and Cooling	Heat pump (water-water)	Hydronic	Grey-box	Not specified
[35]	Residential (single family house)	Austria	Demand response, DHW, load shifting	Resistive heating elements (electrical power)	Not specified	Grey-box	Up to 30% cost savings
[39]	Residential (multi-family) and office buildings	Poland	Heating	District heating	Hydronic	Simplified	Average of 15% energy savings
[42]	Residential (multi-family) and office buildings	Poland	Heating	District heating	Hydronic radiators	Simplified	16.5 energy savings
[41]	Residential (multi-family) and commercial buildings	Poland	Heating	District heating	Hydronic	Not specified	Average energy savings: 13.4% (residential), 10.7% (public)
[37]	Office building	Norway	Heating	District heating	hydronic	Black-box	Not specified
[45]	Office building	China	Cooling	Cooling towers and chillers	Hydronic	Black-box	Energy savings: 5.59%
[46]	Office building	Croatia	Heating and cooling	Not specified	Hydronic and forced air (Air coils)	Grey-box	Reduction of 26% for building operation cost
[24]	Office building	USA	HVAC, thermal comfort	Heat pump (air/air), cooling tower, AHU	Forced air	Grey-box	40% energy saving in HVAC energy
[40]	Office building	Norway	Heating	District heating, electrical panels	Hydronic, electricity	Grey-box	Energy savings of 33%
[47]	Office building	China	Heating	Ground source heat pump	Hydronic	Not specified	9.9–11% energy savings
[48]	Office building	Greece	Space heating and DHW	Multi-source heat pump, PVT collectors	Hydronic	White-box	7% reduction in heat pump electricity consumption
[49]	Office building	Belgium	Heating, cooling	Ground source heat pump, TABS, AHU	Hydronic	White-box	Energy savings: 53.5%
[50]	Commercial building	China	Energy consumption prediction, energy allocation	Natural gas, Renewable energy	Electricity, Water,	Black-box	25% total energy savings
[51]	Commercial building	Canada	Heating	Natural gas boilers and electric boilers	Hydronic	Black-box	Approximately 22% reduction in natural gas consumption
[25]	Commercial building	USA	Cooling, load shifting	PV, batteries, Rooftop chillers	Forced air	Grey-box	12% annual electricity cost
[52]	Commercial building	USA	Demand response, load shifting, domestic hot water heating	PV, battery storage	Not specified	Not specified	Not specified
[53]	Hospital	China	Cooling (only chiller plant)	Chillers and cooling towers	Hydronic	Black-box	2.70% electricity energy saving
[44]	Hospital	Italy	Heating, cooling, electricity	CHP, absorption chiller, electric chillers, steam boilers	Not specified	Not specified	Reduced electricity purchased by 21.49%
[54]	Greenhouse (Industrial)	South Korea	Ventilation, IAQ	Natural ventilation (window openings)	Forced air	Black-box	Not specified
[55]	Factory building (Industrial)	South Korea	HVAC systems	Chillers for AHU	Forced air	Black-box	Approximately 31.7% electricity savings
[56]	Factory building (Industrial)	South Korea	Heating and thermal comfort	Steam boilers serving unit heaters	Forced air (Unit heater operating on hydronic)	Black-box	56.3% energy savings
[57]	Educational buildings	Belgium	Heating, ventilation IAQ	AHU, wood pellet boiler	Forced air	Grey-box	10–44% electric energy use reduction
[58]	Airport Terminal	USA	Cooling, load shifting	Not specified	Hydronic	Black-box	7% energy savings
[59]	Airport Terminal	China	Cooling	Heat pump	Air-forced	Grey-box	24.5% reduction in energy consumption, discomfort time reduced from 70.2% to 5.7%
[60]	Not specified (District heating)	Canada	District heating	Natural gas boilers	Hydronic	Black-box	1.3%–2.8% GHG emission reduction

(*continued on next page*)

Table 1 (*continued*)

Article	Building Type	Country	Area of Application	Energy system	Energy distribution	Building Model Method	Savings
[61]	Not specified (District heating)	Austria	District Heating	Biomass boilers, natural gas boilers, CHP, Thermal energy storage	Hydronic	Black box	35% reduction in CO2 emissions
[62]	Not specified	China	Heating	Solar energy, electric boilers and TES	Hydronic	Not specified	Not specified

Table 2
Number of studies by building type.

Building type	Number of studies
Residential (single family house)	24
Multi-family residential building	5
Multi-purpose (Residential and office building)	3
Multi-purpose (Residential and commercial building)	1
Office building	25
Commercial building	11
Educational building	10
Laboratory building	5
Industrial building	4
Hospital	4
Airport Terminal	2
Not specified	18

building systems (TABS). By adjusting setpoint values in the Building Management System (BMS), the study showed a 30% reduction in heating demand compared to rule-based control [96]. Another German study used a grey-box MPC for HVAC systems and solar shading in an office space, resulting in significant energy saving [97]. A study in the US implemented a grey-box MPC in an office building, reporting a 40% reduction in energy consumption [24]. In Croatia, a field demonstration of MPC in an occupied skyscraper managed heating and cooling using both supervisory and low level control [46].

Office building MPC demonstrations cover several methods and objectives. Using grey-box RC models, MPC for hydronic and air handling systems has achieved about 15–35% energy savings in Germany, Italy, and the United States [96,98,99]. Cooling focused applications are reported in Japan, where the emphasis is on air conditioning control [100]. Black box and ANN approaches show substantial savings in a Canadian office, with heating reductions of 27–42% and cooling reductions of 18–33% [101]. Targeting specific equipment and services, similar black box approaches have been implemented for smart electric heaters in Italy, solar air heating in Iran, and cooling control in Greece [102–104]. Physics-based MPC integrating GSHP and TABS has achieved deep savings in a Belgian office building, achieving a 53.5% reduction in energy use [49]. MPC integrated with PV and battery storage has also been demonstrated in an office context, improving PV self-consumption by 19.5% and building self sufficiency by 10.6% in Singapore [105]. Additional deployments include GSHP based heating control in a Chinese office with approximately 10–11% energy savings [47], and a district-heating office in Norway that reported operational improvements without a quantified savings figure [37].

Residential buildings and small scale applications

Residential buildings often implement MPC, as seen in Section 3.1.1.

One study presented a forecast control system for heating in both existing and new buildings with hydronic heating systems [39]. Using a simple model based on weather forecasts, this system was successfully deployed in several buildings in Poland, revealing average energy savings of 13.4% in residential buildings and 10.7% in public buildings [41]. Another study in the US applied MPC to manage a multi-source energy system in a modified single family house, achieving an 18.7% reduction in energy use [15]. Additionally, a humidity-aware MPC for residential air conditioning showed that effective humidity control could enhance comfort and energy efficiency [31]. In Canada, a highly glazed building used MPC to control hydronic floor heating and cooling, highlighting that simplified solar-gain representations and limited irradiance forecasting can drive systematic overheating or overcooling episodes in highly glazed buildings [36]. A separate study presents predictive controls for a heat pump with thermal storage in a single family house with a photovoltaic rooftop system [34].

Educational and healthcare facilities

MPC has also been applied in educational and healthcare buildings to optimise HVAC operations and maintain occupant thermal comfort. In Italy, MPC controlled heating and cooling in a hospital, demonstrating its effectiveness in managing energy use in a large-scale facility [44]. It optimised energy usage across different energy carriers and operated separately from the BMS to ensure service continuity. A dedicated on-site computer with a resilient power supply ran the MPC, enabling real time communication with the BMS. The optimisation objective was to utilise the hospital's thermal power stations efficiently, focusing on energy usage rather than room temperatures or ventilation fan speeds.

In school buildings, MPC was deployed in six US classrooms to manage HVAC systems during both heating and cooling seasons, achieving a 30% reduction in HVAC peak power and shifting 16% of the load from high price peak periods to lower price periods while maintaining thermal comfort [106]. Another study examined simulations and field implementation of an MPC strategy in a Canadian school using heuristic MPC with radiant floor heating [107]. Two grey-box RC models were used: one for rooms with radiant floor heating and one for rooms with convective heating. The proposed strategy achieved an average of 44% electricity cost savings by shifting load from on peak to off peak periods. Additional European deployments report similar findings. In Belgium, a grey-box MPC coordinated an AHU and a wood pellet boiler, reducing electric energy use by 10–44% [57]. In Denmark, a grey-box MPC focused on heating, ventilation, and IAQ, reporting operational improvements without whole building savings [108].

Industrial- and commercial buildings

Industrial buildings often involve complex energy dynamics, making MPC applications less common. A study in South Korea used a black box MPC with an Artificial Neural Network (ANN) based lumped model to control 12 heating units in a factory, achieving 56.3% energy savings over three weeks while maintaining indoor thermal comfort [56]. In the UK, MPC controlled the HVAC system in a food retail building using a grey-box RC model [109]. The MPC focused on economic optimisation but could also minimise GHG emissions. Due to the high trading intensity, the building experienced significant heat loss, limiting the flexibility provided by MPC. To fully realise the benefits, additional thermal storage was deemed necessary. The case study showed modest cost savings when optimised for cost and negligible savings when minimising GHG emissions. The poor results were attributed to the building's lack of thermal storage and its ventilation based heating system. A study in China used a black-box Long Short Term Memory (LSTM) model to optimise energy management in a multipurpose building, including a shopping mall, offices, entertainment facilities, and a hotel [50]. The author suggests a framework for designing and optimising property energy management systems, which was used in the field study. MPC has also facilitated energy usage shifts in dual fuel heating systems, such as

in a US commercial building where it reduced reliance on a gas fired furnace [110]. Similarly, in Canada, MPC lowered natural gas consumption in an institutional building by 22.2% [51].

As shown in the presented examples, various studies, regardless of building types and locations, considering different case specific elements. Examining correlative factors, geographical location could influence the application area and objectives of the studies. For instance, in Canada, all six implementation studies focus on heating, while in Singapore, Morocco, and Mauritius, all four studies concentrate on cooling. Beyond this obvious climate related correlation, no clear patterns between geographical location and MPC application are found. Regarding building types, the reviewing of the articles indicate that building types could affect the choice of building model being used. For example, of the 14 studies implementing MPC in single family residential buildings, 86% uses a grey-box approach, and 14% utilise a black-box model. For other buildings, these percentages are 50% and 33%, respectively. This is depicted in Fig. 6.

One probable reason for this observation could be that single family residential buildings are generally smaller, making it less challenging to create a detailed mathematical model of the building compared to larger, more complex building types.

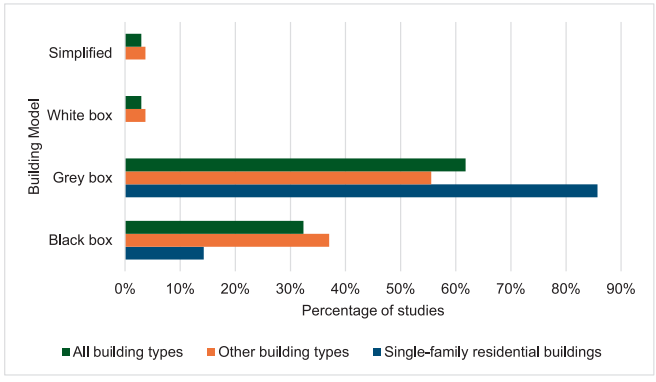


Fig. 6. Distribution of building model being used for single family residential, and other building types.

Energy sources and carriers

The configuration, operation and outcomes of MPC depend on the building's energy system and the controller's optimisation objective. Fig. 7 reports the percentage of studies within three categories. Panel one summarises different optimisation objectives. Panel two highlights the different energy sources, including combined heat and power, photovoltaics, gas boiler, district heating, and grid electricity. Panel three presents the energy carriers that deliver thermal energy to occupied spaces, including air via ventilation, hydronic water distribution, and electricity for direct electric heating. Note that percentages are computed within each aspect, where studies that do not report an aspect are excluded. Also note that a single study can belong to multiple categories within an aspect.

As Fig. 7 suggests, assessing heating using a hydronic system is the most common, likely due to their flexibility in energy sources, including district heating, ground source heat pumps (GSHP), gas furnaces, solar thermal, or combinations of these. Various configurations exist regarding energy carriers and producers, but hydronic systems are more prevalent for heating, whereas air based solutions are more frequently used for cooling. Although less common, there are examples of all electrical systems and climatization control through ventilation air. For all electrical systems, combinations with air to air heat pumps, PV panels, and electrical panels are observed [15,34,52,62]. Energy systems combining different energy carriers are also present. For instance, [111] investigates energy flexibility using a mix of a heat pump, PV

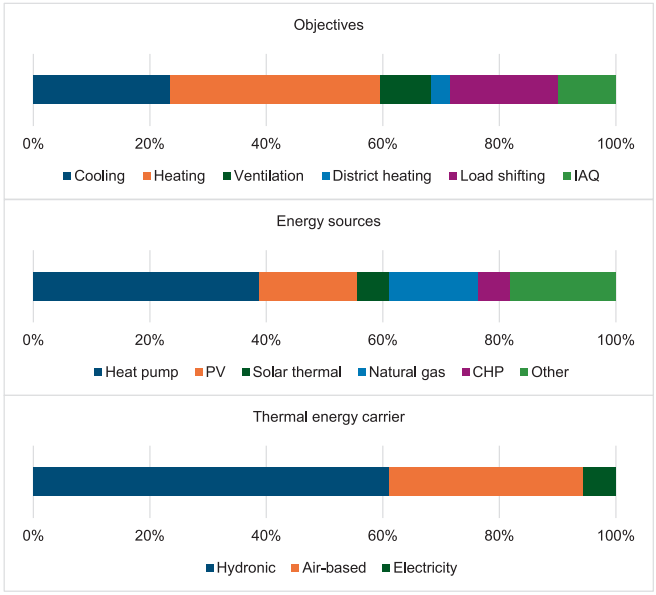


Fig. 7. Percent distribution of studies by objective, energy source and energy carrier. Panel one shows the optimisation objectives addressed, n = 183. Panel two groups the energy sources that supply the system, n = 72. Panel three reports the energy carriers that deliver thermal energy, n = 90.

panels, an electric vehicle (EV), and a battery. Similarly, studies have explored energy flexibility using Thermally Activated Building Systems (TABS) [96,112].

Models used and their complexity

An important aspect distinguishing studies on MPC is the type of models used, classified as black, white, or grey. Each category exhibits differences in complexity, accuracy, simplicity, and availability. MPC for building control often requires models that go beyond the building's energy profile, incorporating factors such as envelope properties, heat capacity, occupancy patterns, solar radiation, and boreholes. Consequently, the control-oriented model used in MPC operations typically consists of multiple sub models. A critical consideration is the robustness of these models to ensure applicability in real-world scenarios. This section highlights the variety of models used in the included studies and discusses their impact on the results.

Black-box models

Of the 75 studies implementing MPC, 23 utilise black-box models. These studies rely on machine learning techniques to predict the energy and temperature profiles of the buildings. For instance, in a South Korean factory building, an ANN was used to model heating demand [56]. The building was divided into ten work zones, with the ANN determining the operating status of heating units to minimise energy consumption. Using two inputs in indoor and outdoor temperature, and one output for the heater operation status (ON/OFF) the approach resulted in a 56.3% energy savings over three weeks compared to the existing manual control. For cooling, the same author used a similar method to operate chillers in an ice cream factory, achieving a 31.7% electricity savings compared to rule-based control [55].

Additionally, ANNs have been applied for cooling via a Variable Air Volume (VAV) ventilation system using six input variables [113] and for cooling an office room with a heat pump and photovoltaic system using five input variables [63]. A black-box, machine learning occupancy forecast (LSTM) coupled with MPC controlled a standalone ventilation system in a laboratory setting, maintaining indoor air quality and reducing energy use by about 16% compared to ON/OFF control [114].

While ANN is the most commonly used method for black-box

modelling in the included studies, it is not the only approach. For instance, [51] examines five different machine learning methods for MPC modelling and implementation in an institutional building, with Gauss Process Regression (GPR) deemed the best fit. Using this non parametric black-box model, heating demand, electric baseload, and natural gas consumption were predicted to reduce energy usage for heating and natural gas. Implementing this methodology in a case building in Canada resulted in a 4.3% reduction in energy use and a 22.2% reduction in natural gas consumption. Additionally, [45] utilised an ensemble prediction in a cloud edge terminal platform, employing a fuzzy matching black-box model to train multiple models under different conditions to predict the cooling load of a high rise office building in China, achieving a 5.59% reduction in energy consumption.

Grey-box models

Grey-box models are the most commonly used, with 39 out of the 75 implementation studies utilising them. These models often include an RC equivalent for the building system. While an RC representation of a building could be entirely based on physical relationships, it is common to include parameters derived from training data, making it a grey-box model. For instance, a study implemented MPC in a section of a large office building in Germany using a 7R4C model created in Modelica [96]. The semi physical model included free parameters calibrated using measured data from previous years of operation, and the controller achieved a 30% reduction in energy demand compared with rule-based control during a three-month test period.

A similar method was used in the real-world implementation and evaluation of a physical based Modelica building model presented in [97]. Data from construction plans were used to make an initial building model, which was then calibrated using iterative, nonlinear parameter estimation. The study investigated radiator and floor heating and shading in a laboratory space in Germany over 20 days, with heating output controlled directly from the MPC output, resulting in significant energy savings.

Other examples include [51], which used an MPC adopting a linear state space model, where a thermal RC model (6R3C) and AI were used to reduce gas consumption. [115] presents an experimental study of MPC for air conditioning and mechanical ventilation systems in Singapore, evaluating both thermal comfort and cooling.

In Switzerland, a grey-box MPC was developed for a multi use building utilising ground source heat pumps (GSHP) and district heating [95]. This study used a black-box model (ANN) for the building's energy profile and a physical white-box model for thermal storage and the heat pump, allowing for 13.4% flexibility in electricity consumption due to the buffer provided by water storage tanks.

Other implementations broaden the scope of MPC across building types and services. Grey-box control of AC systems for cooling and load shifting in residential and commercial settings reports cooling cost reductions of about 22–27% and energy cost savings of about 4–9%, with residential cooling energy reduced by about 9–18% [116–118]. Cost oriented heating and cooling with natural gas systems yield 6–12% demand cost savings [119]. TES-integrated hydronic systems in educational and institutional contexts enable load shifting and flexibility, with outcomes reported qualitatively [120,121]. Residential heat pump heating under grey-box MPC delivers about 10–11% energy savings [122].

White-box models

Completely white-box models are generally difficult to adjust into a control-oriented model, and it is often stated that white-box models are unsuitable for control purposes. However, the included studies show that there have been successful implementations of MPC using white-box models. For example, [49] demonstrates a detailed method of cloud based MPC implementation, field tested on an office building with a ground source heat pump and thermally activated building structures (GEOTABS) in Belgium. Another study presented the use of white-box models for a heat pump based energy system for space heating and hot water production [48]. While these examples demonstrate that white-box models can be used for control purposes, they still present challenges, especially for scalability. The main challenge mentioned in the latter study is the significant effort required to develop sufficiently accurate white-box models of the system components, reducing the replicability of the algorithm in buildings without identical system components [48]. White-box implementation is also troublesome due to dynamic disturbances like occupant behaviour, as well as limited point measurements [123].

Several simulation studies also employ white-box models across different contexts. In single family configurations in Germany, a white-box approach reported up to about 30% space heating demand reduction [72]. In an Argentinian hospital, white-box MPC for HVAC and demand response improved thermal comfort by up to 47% and reduced electricity costs by 23% compared to a strategy without peak clipping [124]. A single family house in Italy using an air source multi split system achieved 28% cooling cost savings under white-box MPC [125]. An educational building in Norway reported about 3.4% weekly energy cost savings with hydronic systems and load shifting [126]. Additional studies demonstrate white-box MPC for solar collector and air to water heat pump hydronic systems in a public building [127], district heating ventilation and IAQ control in Sweden [128], and residential hydronic heat pump operation in Germany [71].

Sensors and data

To automate building operation, the controller must know the building state. Key signals include indoor temperature, CO_2 and occupancy. In rule-based control these measurements simply trigger responses, for example more heat when the measured temperature is low. In MPC the same data also support model identification and validation. Effective control further requires actuation, such as valves, fans or heating and cooling setpoints. Together, sensing and actuation define what is feasible in a given building, and newer buildings usually offer richer sensing and more precise control than older stock. These interactions have become more accessible through the IoT, defined as the network of physical objects with embedded technology to communicate and sense or interact with their internal states or the external environment [129]. By this definition most medium and large buildings already expose IoT through their BMS, sensors and HVAC equipment, while IoT also extends connectivity to smaller assets and domestic devices, which enables MPC in small housing and retrofits without investment in a full BMS.

From the literature reviewed, most studies utilise IoT to some degree. Whether it is the focus of the study varies, with some papers highlighting it while others do not. While examples of studies using IoT have already been discussed in Section 3.1.1. Another example is given in [98], which presents an IoT based architecture for MPC of HVAC systems in a campus building in Italy, relying on AI for model development to reduce gas consumption. Sensors on HVAC modules send measurements to a multi protocol gateway and then to a database. A control unit retrieves these measurements and weather forecasts via an API, runs the MPC algorithm, and issues commands back to the actuators through the same gateway. Users access dashboards connected to the database so that preferences and system status are visible and can inform operation.

A study investigate implements a low cost MPC for a building heating system, using data from IoT and dynamic pricing, locally deployed on Raspberry Pi hardware via WiFi [24]. This study is conducted in a test pod in the UK, where the MPC control the output of an electric heater while at the same time investigate how PCM influence the system.

[130] presents an innovative study on data-driven predictive control (DPC) for building energy management under a four tier building energy IoT architecture. A cloud based SCADA building energy management system framework is developed for standardising communication protocols and data formats, favourable for advanced control strategies

implementation. The method is validated with simulations and is thus not implemented in an operated building. In their work [28], the authors developed an MPC agent and implemented it on an online cloud based platform using smart home thermostats. The MPC agent uses a grey-box model to capture home thermal dynamics and a node sensor for interior wall surface temperature measurement. Field tests on nine homes over four months demonstrated its effectiveness.

Monitored parameters and IoT trends

Across the reviewed implementations, the most common monitored parameters are consistent and pragmatic. Zone level indoor temperature is nearly universal, serving as both the primary comfort variable and the basis for model identification. Forecasted outdoor air temperature is likewise standard, as it dominates the disturbance profile and enables anticipatory control [39,41,44]. When latent heat loads or indoor air quality are considered, studies typically add relative humidity and CO_2 measurements, as in humidity-aware MPC and classroom ventilation control [31,57]. When cost or flexibility is in focus, metered energy and tariff or demand-response signals are added. If PV or batteries are present, PV output and battery state of charge are included [25,29,34,52]. Solar gains and shading are sometimes measured or modelled to improve envelope heat balance prediction [97].

Two complementary trends appear. First, IoT is broadening data availability. Smart thermostats, wireless sensors and multi protocol gateways expand room level coverage. They also increase measurement frequency and simplify access to weather, tariff, and DER data. As a result, MPC becomes more feasible in homes without a BMS, where pilots have used smart thermostats and wireless sensors to provide essential signals and actuation while avoiding rewiring and enabling low cost controllers [15,28,29,31]. In both homes and larger buildings, IoT can shorten the time required for setup, integration, and early testing of MPC. It also makes it easier to add additional signals when needed, for example for economic MPC or flexibility services [52,98].

Second, several deployments show that effective control does not require dense sensing. Grey-box RC models can estimate unmeasured envelope states from limited measurements [97], and proxy variables, such as ventilation extract temperature as a stand in for zone temperature, have also been used [43]. Black-box predictors with few inputs have been applied to heating and cooling plants in industrial and institutional settings [51,55,56]. At portfolio scale, simplified, weather-driven supervisory heating control with limited instrumentation has achieved double digit savings in hydronic buildings.

In practice, this suggests a pragmatic sensing strategy. Start from a compact core, namely zone indoor temperature, outdoor temperature and forecast, and one or two plant signals indicative of delivered or input energy. Then add only those measurements that materially improve the objective at hand. Humidity and CO_2 are natural additions for comfort and IAQ, while solar or shading, sub metering and tariff or demand response feeds are high leverage for economic or flexibility goals [28,31,40,43]. As IoT enlarges the potential dataset, many controllers can still remain simple. Where richer data are available or required, black-box components can be introduced selectively, provided sufficient training data and careful validation are in place. Using a small set of measurements with surrogate models, and adding only targeted IoT signals, keeps commissioning effort low while preserving field performance.

Approaches for scalability

MPC for building control can take various approaches for modelling, prediction, and control. One major obstacle for large scale implementation is the need for a detailed building model for accurate predictions. There are examples where this issue is mitigated by using less detailed, simpler models, as discussed in Section 3.2.2. Another approach is to introduce model archetypes, which are pre developed models for different scenarios, reducing the time spent on model development for each building. For instance, [131] created three archetypes of MPC for heating with heat pumps in residential buildings, dependent on the heat sink type. This methodology enables applying MPC archetypes to different case studies, including data for each building's dynamics. Another example is found in [132], where control-oriented archetype models of different RC-model configurations are developed for typical Canadian zones and building configurations, while CO_2 prediction models for educational buildings are shown in [133]. Although these archetypes were validated through simulations, they have not yet been deployed in real-world operations. Although different archetypes could reduce the amount of time for model development, finding a "one size fits all" archetype has proven to be challenging. While simulation studies are showing promising results, testing and implementing the suggested archetypes should be done in a more extensive manner.

Other techniques for easier implementation include rule extraction, as presented in [101]. This study involved a field implementation in an office in a Canada to control heating and cooling while considering thermal comfort. The results showed that implementing MPC using machine learning state space models and rule extraction led to 42% and 27% energy savings for cooling, and 18% and 33% for heating.

While these approaches to improve scalability are promising, quantitative benchmarking remains limited because deployment effort is rarely reported in field demonstrations. This limitation has been quantified in the literature. Only 13 of 104 field-demonstration papers report deployment, operation, or maintenance costs, which constrains direct quantitative benchmarking across studies and highlights the need for more consistent reporting of deployment effort [134].

Performance of MPC

As the studies show, MPC with various models and objectives has been successfully implemented using diverse techniques and methods. This section examines the outcomes of the included studies. An intuitive question is whether there is any correlation between the model used, implementation technique, application area, and reported savings. However, cross-study comparability is limited because context and baselines are not reported consistently, including building characteristics, occupancy, evaluation periods, and the definition of the reference controller. Savings are also reported for different scopes, for example a single subsystem versus whole building energy use. Additionally, there are no standard methods for calculating savings, and studies vary in whether they report energy, cost, emissions, self-sufficiency, or flexibility metrics. These factors constrain the use of a formal *meta*-analysis and limit the interpretability of pooled effect sizes.

Nevertheless, a concise quantitative overview provides a common reference point for interpreting reported savings and complements the qualitative discussion of implementation patterns. We therefore report descriptive statistics of the reported energy and cost savings for the subset of studies that provide extractable values. Savings are expressed as percentages relative to each study's stated baseline. When studies reported savings as ranges, we used the midpoint of the reported range as a representative value. Table 3 summarises the results for all implementation studies combined and separately for full-scale implementation studies and for studies conducted in parts of a building or in laboratory settings.

Across the studies that reported energy savings, the mean energy saving was 20.4% and the median was 16.5% ($n = 31$), with substantial dispersion and a wide range of outcomes. Cost savings show a similar central tendency, with mean 26.1% and median 26.0% ($n = 15$). The smaller sample size for cost reflects that fewer studies quantify monetary outcomes in a comparable way. When grouped by implementation scale, full-scale implementation studies report slightly higher central tendency than the overall set, while variability remains high and the number of cost-saving data points is limited. Studies conducted in parts of a building or in laboratory settings show a lower mean energy saving,

Table 3
Descriptive statistics of reported energy and cost savings by implementation scale. Savings are expressed as percentages relative to each study's stated baseline.

Across all implementation studies						
Metric	n	Mean	Median	Standard deviation	Min	Max
Energy savings	31	20.4	16.5	14.3	1.70	56.3
Cost savings	15	26.1	26.0	13.0	6.14	55.0
Across studies with implementation in parts of a building or laboratory settings						
Metric	n	Mean	Median	Standard deviation	Min	Max
Energy savings	14	17.6	16.0	11.7	1.70	40.4
Cost savings	8	22.5	22.2	12.0	6.14	44.0
Across studies with full-scale implementation						
Metric	n	Mean	Median	Standard deviation	Min	Max
Energy savings	17	22.7	19.0	16.1	2.7	56.3
Cost savings	7	30.1	27.0	13.8	12.0	55.0

although median values are similar to the overall sample, which may reflect the narrower system scope and fewer opportunities for whole-building optimisation. These patterns should be interpreted cautiously given cross-study variation in baselines, scope, and reporting practices.

Fig. 8 complements Table 3 by visualising the spread of reported energy and cost savings and highlighting extreme values across studies.

Taken together, Table 3 and Fig. 8 show wide dispersion in reported performance, reaching maxima of 56.3% for energy [56] and 55% for cost [29], with a minima of 1.70% and 6.14%, respectively. This variability is consistent with the practical issues discussed earlier, notably inconsistent baselines, differences in HVAC scope under MPC, varying evaluation periods, and site-specific drivers such as envelope quality, weather, and occupancy, all of which shape reported savings and limit direct cross-study comparability.

Despite this cross-study variation, an interesting pattern is that higher savings are more often reported in single family residential buildings and in older buildings [28,29,40,56], where baseline controls and retrofit constraints may leave greater optimisation potential.

Conclusion

This review underscores the significant potential of MPC to enhance energy efficiency and operational performance across a wide range of building types. The key findings are as follows:

MPC implementation varies significantly in methodological approaches, building types, climatic conditions, and control objectives.

MPC is most frequently applied in hydronic heating systems due to their operational flexibility and ability to integrate multiple energy sources.

Grey-box models are the most widely used modelling approach, as they combine physical insight with data-driven parameter estimation, while black-box and white-box models each present trade-offs in data requirements, interpretability, and scalability.

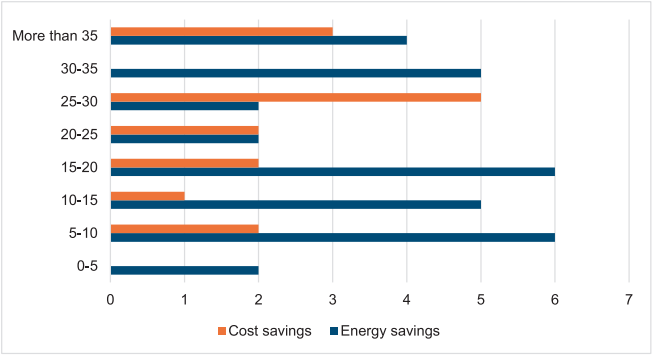


Fig. 8. Reported energy and cost savings across the included studies.

Practical challenges include communication interoperability, sensor accuracy, connectivity, data resolution, and model reliability, while lower-cost MPC solutions based on existing infrastructure and simplified models show promise for wider adoption.

Scalability remains a critical challenge due to case-specific implementations, with differences in building characteristics, control systems, climate, occupancy, and performance metrics limiting direct comparability across studies.

Greater emphasis on replicating successful methodologies across multiple buildings is required to demonstrate scalability and robustness of MPC in real-world applications.

A further implication for sustainable smart-city development is that MPC should be considered as part of an integrated strategy rather than a stand-alone solution. Energy efficient building design reduces heating and cooling demand, while on-site renewable energy systems such as solar PV can supply part of the electric power load. In conclusion, despite challenges related to scalability, MPC has been successfully applied across diverse building types and energy systems, supporting its potential as a key technology for more efficient and flexible building operation.

Future outlook and emerging technologies

The evidence synthesised in this review has implications for theory, managerial practice and policy decisions. These implications are outlined below to support both research and implementation.

Scalable deployment, interoperability, and digital twins

A recurring barrier within the included studies was interoperability across devices and systems. This includes limited access to setpoints, inconsistent sensor tagging, and vendor-specific constraints [15,32,37]. Semantic standards from Building Information Modelling (BIM), such as the Industry Foundation Classes (IFC) provide a common machine-readable vocabulary for points, equipment, and zones. Combined with open interfaces such as BACnet and MQTT, and supported by standard data models, they enable automated discovery, mapping and validation of the data pathways required for MPC. This support faster commissioning and more repeatable deployments. Digital twins offer a complementary route to scale MPC implementation. It can also combine physics-based and data-driven components. In addition, it enables parallel testing so control strategies can be tuned before deployment.

At multi-building scale, templated digital twins aligned with building archetypes (Section 3.6) allow reuse of model structures while adapting parameters to each site. They also support monitoring, fault detection, and measurement and verification to maintain performance over time.

Learning enabled and adaptive MPC

Grey-box models dominate practical implementations, particularly in residential buildings, as they balance physical insight with tractable parameter identification. Learning enabled approaches can further reduce engineering effort by automating model discovery, speeding calibration, and maintaining accuracy under system operation, are supported by advances improvements in embedded optimisation and fast solvers [11,20]. Online parameter estimation and transfer learning can initialise models from archetypes, and adapt them to new buildings having limited data. This reduces on-site calibration effort [12,23]. In multi-zone and usage-dependent buildings, *meta*-learning may enable faster updates when occupancy patterns, building properties, or equipment change.

Learning also supports occupant-centric control. They allow adaptation of comfort constraints and preferences during operation, while respecting actuator limits. In parallel, uncertainty-aware formulations remain practical enablers. Robust, stochastic, and economic MPC can

reduce forecast errors and co-optimise cost and efficiency [17,18]. This can be embedded in hierarchical or distributed architectures, supports grid-interactive operation under dynamic tariffs and flexibility markets [25,60]. Edge-cloud co-design and hierarchical control reduce real-time computational burden. At the same time, they preserve the ability to optimise economic performance at higher levels.

Alternative control paradigms

Although MPC is the most established advanced control approach in real buildings, alternative and complementary methods are emerging. Reinforcement learning has shown promising results in simulation studies for HVAC control, demand response, and multi-objective operation. It can learn control policies without an explicit plant model. However, evidence from occupied buildings remains limited. Key barriers include data efficiency, explicit handling of comfort and actuator constraints, interpretability, and integration with existing BMS. Recent work on safe and constrained learning, offline training using historical data, and model-based reinforcement learning aims to address these limitations. However, field validation remains limited [135].

Scaling demonstrations and addressing first use challenges

Follow-up demonstrations are expected to build on initial pilots by addressing first-use barriers such as integration effort, setup cost, and commissioning time. As experience accumulates, the focus is likely to shift towards scalability. Subsequent deployments can be made more streamlined and cost-effective through standardised workflows and reusable methods.

Industry uptake and policy levers

A persistent gap is the limited involvement of the automation industry and other key stakeholders. In many sectors, technology adoption is industry driven. However, MPC in buildings remains largely research led, with only a limited number of commercial solutions available. The potential for energy and operational cost savings suggests that industry engagement may increase. This is particularly likely as policy targets become more stringent and dynamic pricing mechanisms become more widespread. In parallel, enabling technologies are reducing deployment barriers.

Targeted incentives can further support adoption as dynamic tariffs and flexibility markets expand. Early-stage subsidies, performance-based payments, and utility programmes can lower perceived risk and encourage uptake by vendors and building owners. Where flexibility is valued, tariff structures and procurement mechanisms can reward predictive control by valuing peak reduction and load shifting alongside energy savings. Targeted policy instruments incentives can further support adoption. These include early-stage subsidies, performance-based incentives, and utility-led programmes that reduce perceived risk for vendors and building owners. As flexibility markets expand, tariff structures and procurement mechanisms can reward predictive control by valuing peak load reduction and load shifting alongside energy savings.

Research on practical and interdisciplinary challenges

Practical implementation challenges remain. These include improving robustness in real-world conditions, integrating MPC with existing BMS, and ensuring transferability across building types and climates will be important. Progress is likely to depend on closer collaboration among researchers, software developers, and the construction and operations sectors. Such collaboration is needed to translate prototype systems into reliable operational practice. In smart-city contexts, additional opportunities arise. On-site renewable energy systems, such as solar photovoltaics, and ground-source heat pumps and district heating, can increase the value of predictive operation. This makes MPC particularly relevant for flexible and cost-effective building performance.

Future research — summary

Future research priorities include:

- Portfolio-scale replication studies using standardised measurement and verification frameworks, and a consistent reporting protocol that requires explicit baseline definition to enable cross-study comparability.
- Field testing of archetype-based digital twins across different building types to assess transferability and scalability of control strategies
- Development of online identification and transfer learning methods, alongside evaluation of uncertainty-aware and economic MPC under operational uncertainty.
- Quantification of trade-offs between additional sensing requirements and achieved performance improvements.
- Regular mapping of industry to track progress, identify barriers, and support evidence based policy development.

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CRediT authorship contribution statement

Lars Øgar Rastad: Writing – original draft, Visualization, Software, Methodology, Investigation. **Arnab Chaudhuri:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Habtamu Bayera Madessa:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix I

Table 4
Article- and case buildings country origin.

Country	Article origin	Case building location
United States	25	17
China and Hong Kong	18	10
Italy	14	8
Canada	12	12
Germany	9	12
Belgium	7	6
South Korea	7	6
United Kingdom	6	3
Greece	5	2
Denmark	4	5
France	4	–
Poland	4	4
Sweden	4	4
Switzerland	4	5
Austria	3	3
Singapore	3	2
Spain	3	–
Australia	2	1
Iran	2	1
Morocco and Casablanca	2	2
Netherlands	2	2
Norway	2	3
Argentina	1	1
Croatia	1	1
Finland	1	–
India	1	1
Japan	1	1
Kiona as	1	–
Mauritius	1	1
United Arab Emirates	1	–
Portugal	–	1
Not specified	–	5

Data availability

Data will be made available on request.

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